exploratory tools into core enablers of interplanetary industry and infrastructure development.

9.8 Case Study

9.8.1 Military swarm drones – cooperative autonomy in combat environments

Military swarm drones exemplify the application of distributed autonomy to achieve force-multiplying effects on the battlefield. Unlike single-platform UAVs, swarm systems consist of many relatively inexpensive, networked micro- or small-UAVs that cooperate to accomplish reconnaissance, electronic warfare, decoying, target acquisition, and precision strike missions. Each unit operates with local sensing and decision rules while sharing state information with neighbors; collectively the swarm self-organizes, reassigns tasks, and adapts to loss or interference without a single point of failure.

Operational employment of swarms emphasizes rapid area coverage and resilience. In reconnaissance roles, hundreds of small drones can distribute across a theater to generate high-resolution, multi-perspective sensor coverage, enabling commanders to detect movement, map terrain, and update situational awareness continuously. For offensive operations, swarms can perform saturation or distributed-attack tactics presenting many simultaneous threats to overwhelm air defenses and complicate targeting. Swarms are also used as electronic warfare nodes to create coordinated jamming patterns, and as decoys to draw enemy fire or reveal air-defense positions, preserving higher-value manned assets (Grimal, et al. 2018).

Technically, effective military swarms integrate four elements: robust local autonomy (collision avoidance, formation keeping), reliable inter-agent communication (mesh or relay architectures resilient to jamming),